

Quiz Score Totals

Code:

```
from airflow import DAG # type: ignore

from airflow.providers.standard.operators.python import PythonOperator # type: ignore
from datetime import datetime

def maths_score(**kwargs):
    ti = kwargs['ti']
    ti.xcom_push(key="maths", value=78)
    print("Maths score pushed: 78")

def english_score(**kwargs):
    ti = kwargs['ti']
    ti.xcom_push(key="english", value=85)
    print("English score pushed: 85")

def science_score(**kwargs):
    ti = kwargs['ti']
    ti.xcom_push(key="science", value=62)
    print("Science score pushed: 62")

def calculate_average(**kwargs):
    ti = kwargs['ti']
    maths = ti.xcom_pull(key="maths", task_ids="maths_score")
    english = ti.xcom_pull(key="english", task_ids="english_score")
    science = ti.xcom_pull(key="science", task_ids="science_score")

    average = (maths + english + science) / 3
    if average >= 90:
        grade = "A"
    elif average >= 75:
        grade = "B"
    elif average >= 60:
        grade = "C"
    else:
        grade = "D"
    print(
        f"""
Maths: {maths}\nEnglish: {english}\nScience: {science}\nAverage:
{average}\nGrade: {grade}
        """, flush=True
    )
    return {"average": average, "grade": grade}
```

```

def print_report(**kwargs):
    ti = kwargs['ti']

    maths = ti.xcom_pull(task_ids="maths_score", key="maths")
    english = ti.xcom_pull(task_ids="english_score", key="english")
    science = ti.xcom_pull(task_ids="science_score", key="science")

    result = ti.xcom_pull(task_ids="calculate_average")

    print(
        f"""
----- QUIZ REPORT -----
Maths: {maths}\nEnglish: {english}\nScience: {science}\nAverage:
{result['average']}\nGrade: {result['grade']}
-----
        """, flush=True)
with DAG(
    dag_id="quiz_scores",
    schedule="@once",
    start_date=datetime(2024, 1, 1),
    catchup=False
) as dag:
    maths_score = PythonOperator(
        task_id="maths_score",
        python_callable=maths_score
    )

    english_score = PythonOperator(
        task_id="english_score",
        python_callable=english_score
    )

    science_score = PythonOperator(
        task_id="science_score",
        python_callable=science_score
    )

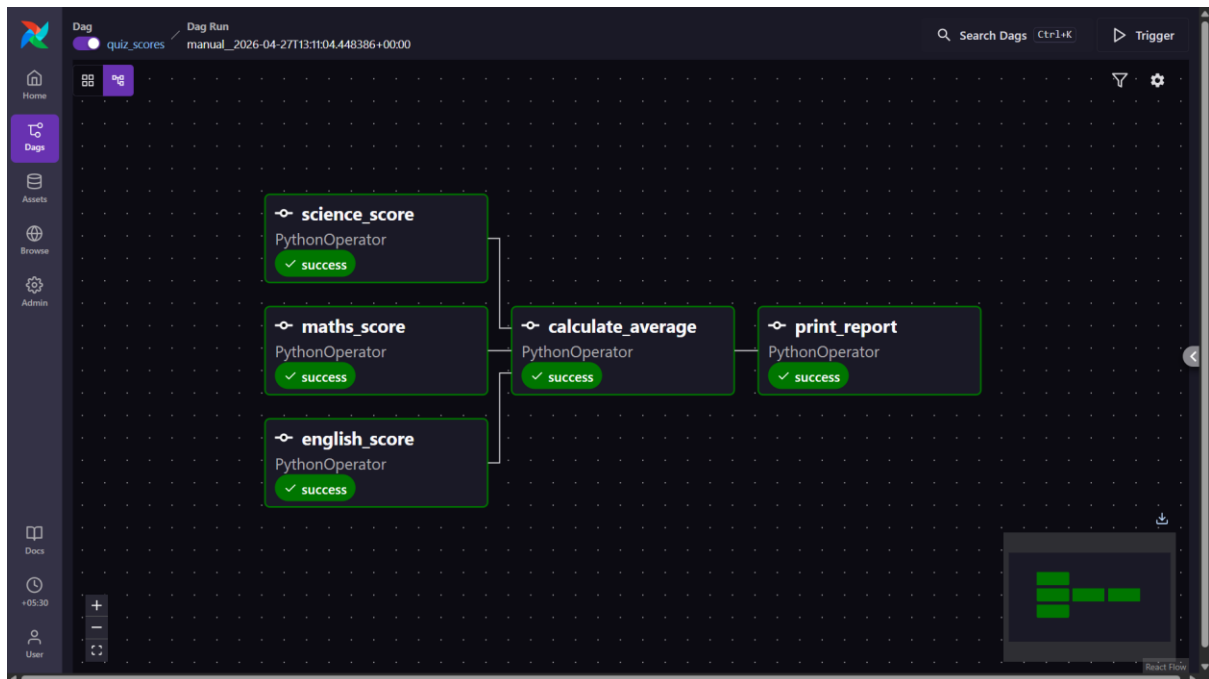
    calculate_average = PythonOperator(
        task_id="calculate_average",
        python_callable=calculate_average,
        trigger_rule="all_done"
    )

    print_report = PythonOperator(
        task_id="print_report",
        python_callable=print_report,
        trigger_rule="all_done"
    )

```

```
[maths_score, english_score, science_score] >> calculate_average >>
print_report # type: ignore
```

1. Graph View — all 5 tasks GREEN, 3 score tasks at same level then calculate then report



2. Admin → XComs page — showing the 3 named XCom entries (maths, english, science keys visible)

The image shows the Airflow XComs page for the 'quiz_scores' DAG. The table displays the following XCom entries:

Key	Dag	Dag Run ID	Run After	Task	Map Index	timestamp	Value
return_value	quiz_scores	manual_2026-04-27T13:04:13.863764+00:00	2026-04-27 18:34:13	calculate_average	-1	2026-04-27 18:34:15	<pre>{ "grade": "B", "average": 75 }</pre>
english	quiz_scores	manual_2026-04-27T13:04:13.863764+00:00	2026-04-27 18:34:13	english_score	-1	2026-04-27 18:34:14	85
maths	quiz_scores	manual_2026-04-27T13:04:13.863764+00:00	2026-04-27 18:34:13	maths_score	-1	2026-04-27 18:34:14	78
science	quiz_scores	manual_2026-04-27T13:04:13.863764+00:00	2026-04-27 18:34:13	science_score	-1	2026-04-27 18:34:14	62

3. Task Log of calculate_average — showing all 3 pulled scores, the average, and the grade

The screenshot displays the Airflow web interface. On the left, a DAG named 'quiz_scores' is shown with three tasks: 'science_score', 'maths_score', and 'english_score', all marked as 'success'. The 'calculate_average' task is highlighted with a blue box. On the right, the 'Task Log' for 'calculate_average' is shown, indicating it is 'Success'. The log details include the operator 'PythonOperator', start date '2026-04-27 18:43:26', end date '2026-04-27 18:43:26', duration '00:00:00.196', and dag version 'v3'. The log messages show the task pulling scores for Maths (78), English (85), and Science (62), calculating an average of 75.0, and determining a grade of 'B'. A red box highlights the log messages from line 7 to 14.

```
Log message source details sources=["/usr/local/airflow/logs/dag_id=quiz_scores/run_id=manual_2026-04-27T13:13:24.316329+00:00"]
2 [2026-04-27 18:43:26] WARNING - Astro managed secrets backend is disabled
3 [2026-04-27 18:43:26] INFO - DAG bundles loaded: dags-folder
4 [2026-04-27 18:43:26] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/quiz_scores.
5 [2026-04-27 18:43:26] INFO - TaskInstance Details: dag_id=quiz_scores task_id=calculate_average dagrun_id=manual_2026-04-27T13:13:24.316329+00:00
6 [2026-04-27 18:43:26] INFO -
7 [2026-04-27 18:43:26] INFO - Done. Returned value was: {'average': 75.0, 'grade': 'B'}
8 [2026-04-27 18:43:26] INFO - Maths: 78
9 [2026-04-27 18:43:26] INFO - English: 85
10 [2026-04-27 18:43:26] INFO - Science: 62
11 [2026-04-27 18:43:26] INFO - Average: 75.0
12 [2026-04-27 18:43:26] INFO - Grade: B
13 [2026-04-27 18:43:26] INFO -
14 [2026-04-27 18:43:26] INFO - Pushing xcom ti=RuntimeTaskInstance(id=UUID('019dcf12-ba6d-7256-8944-9b140c'))
```

4. Task Log of print_report — showing the full formatted report

The screenshot displays the Airflow web interface. On the left, the DAG 'quiz_scores' is shown with two tasks: 'calculate_average' and 'print_report', both marked as 'success'. The 'print_report' task is highlighted with a blue box. On the right, the 'Task Log' for 'print_report' is shown, indicating it is 'Success'. The log details include the operator 'PythonOperator', start date '2026-04-27 18:43:26', end date '2026-04-27 18:43:27', duration '00:00:00.120', and dag version 'v3'. The log messages show the task pulling the average and grade from the previous task and printing a formatted report. A blue box highlights the log messages from line 6 to 15.

```
Log message source details sources=["/usr/local/airflow/logs/dag_id=quiz_scores/run_id=manual_2026-04-27T13:13:24.316329+00:00"]
2 [2026-04-27 18:43:26] WARNING - Astro managed secrets backend is disabled
3 [2026-04-27 18:43:26] INFO - DAG bundles loaded: dags-folder
4 [2026-04-27 18:43:27] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/quiz_scores.
5 [2026-04-27 18:43:27] INFO - TaskInstance Details: dag_id=quiz_scores task_id=print_report dagrun_id=manual_2026-04-27T13:13:24.316329+00:00
6 [2026-04-27 18:43:27] INFO -
7 [2026-04-27 18:43:27] INFO - ----- QUIZ REPORT -----
8 [2026-04-27 18:43:27] INFO - Maths: 78
9 [2026-04-27 18:43:27] INFO - English: 85
10 [2026-04-27 18:43:27] INFO - Science: 62
11 [2026-04-27 18:43:27] INFO - Done. Returned value was: None
12 [2026-04-27 18:43:27] INFO - Average: 75.0
13 [2026-04-27 18:43:27] INFO - Grade: B
14 [2026-04-27 18:43:27] INFO -
15 [2026-04-27 18:43:27] INFO -
```